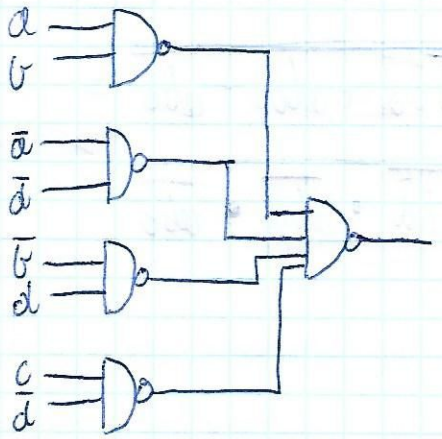
zadanie 1.

Funkcja po przekształceniach: $\overline{a\bar{b}} \cdot \overline{\bar{a}\bar{d}} \cdot \overline{\bar{b}d} \cdot \overline{c\bar{d}}$

Realizacja:

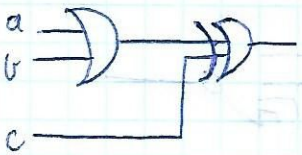


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zadanie 2.

$$f(a,b,c) = \sum m(1,2,4,6) = a\bar{c} + b\bar{c} + \bar{a}\bar{b}c =$$

$$= \bar{c}(a+b) + c(\overline{\bar{a}\bar{b}}) = \bar{c}(a+b) + c(\overline{a+b}) = c \oplus (a+b)$$



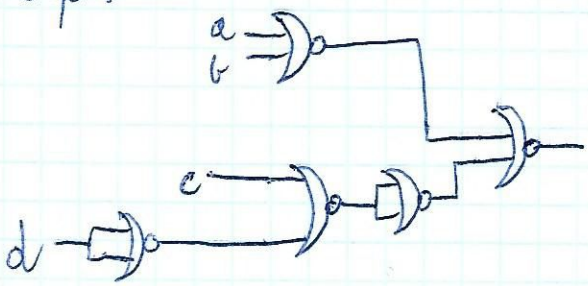
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zadanie 3.

$$f(a,b,c,d) = (a+b)\bar{c}\bar{d} = \overline{\overline{(a+b)\bar{c}\bar{d}}} =$$

$$= \overline{\overline{(a+b)} + \overline{\bar{c}\bar{d}}} = \overline{(\overline{a+b}) + \overline{\overline{c+d}}}$$

Realizacja:



zadanie 4

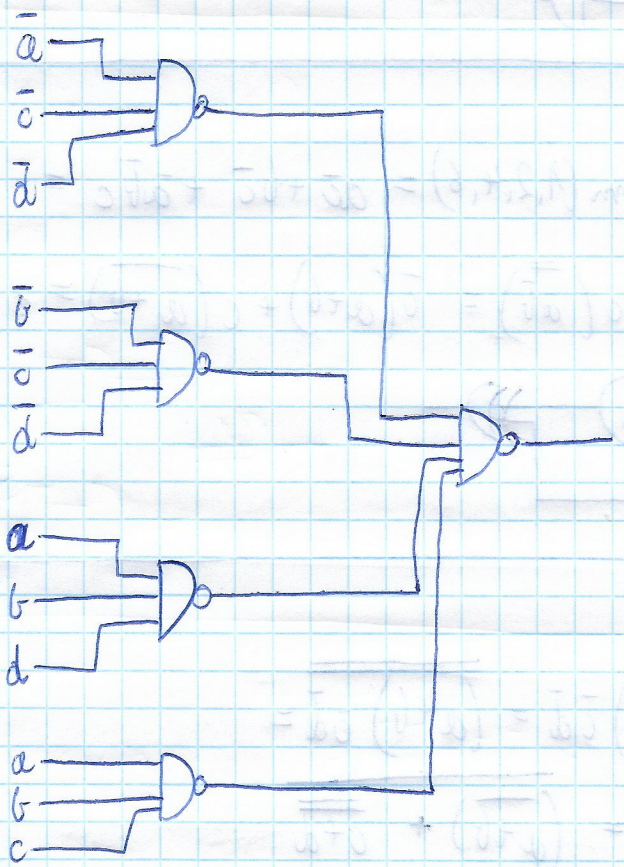
$f(a,b,c,d) = \Sigma m(0,4,8,13,14,15)$

z siatki Karnaugh: $f(a,b,c,d) = \bar{a}\bar{c}\bar{d} + \bar{b}\bar{c}\bar{d} + abd + abc$

a) $f(a,b,c,d) = \overline{\bar{a}\bar{c}\bar{d} + \bar{b}\bar{c}\bar{d} + abd + abc} =$

$= \overline{\bar{a}\bar{c}\bar{d}} \cdot \overline{\bar{b}\bar{c}\bar{d}} \cdot \overline{abd} \cdot \overline{abc}$

Realizacja:

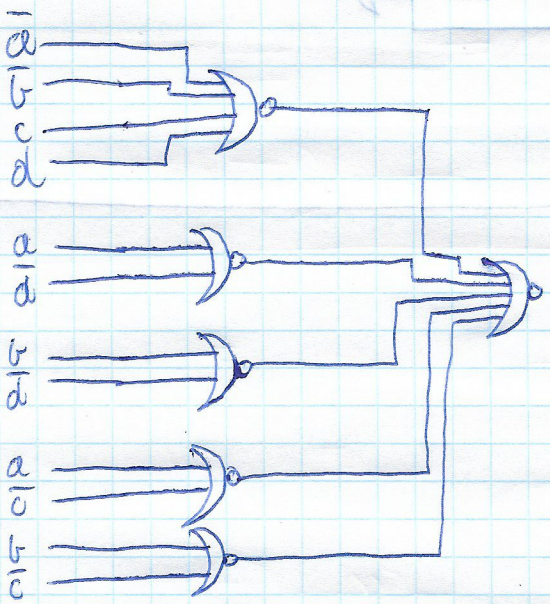


b) z siatki Karnaugh:

dla zer $\overline{(\bar{a}+\bar{b}+c+d)\cdot(a+\bar{d})\cdot(b+\bar{d})\cdot(a+\bar{c})\cdot(b+\bar{c})} =$

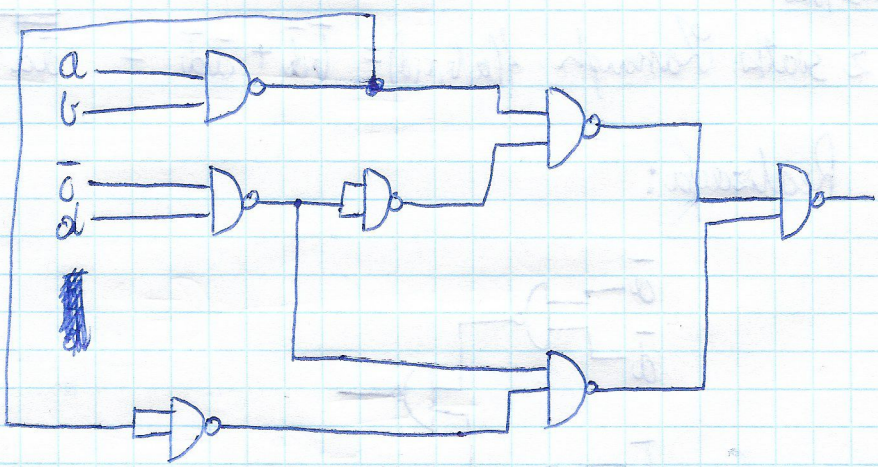
$= \overline{(\bar{a}+\bar{b}+c+d)} + \overline{(a+\bar{d})} + \overline{(b+\bar{d})} + \overline{(a+\bar{c})} + \overline{(b+\bar{c})}$

Realizacja:



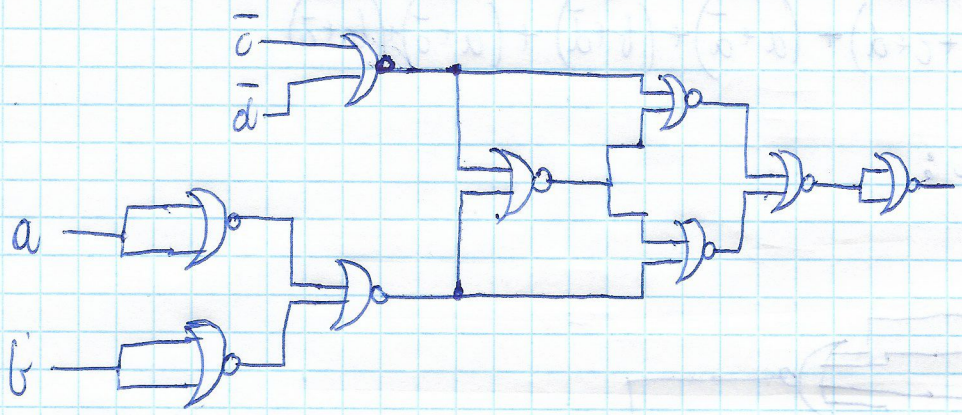
c)

Realizacja:



d)

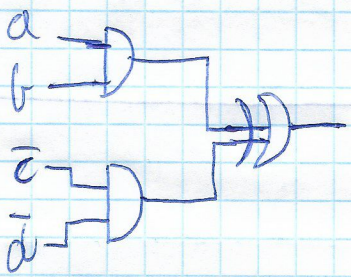
Realizacja:



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e) $f(a,b,c,d) = (ab \oplus \bar{c}\bar{d})$

Realizacja:

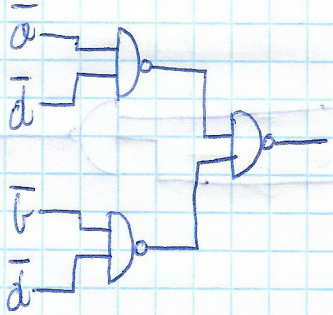


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zadanie 5

z siatki Karnaugh: $f(a,b,c,d) = \bar{b}\bar{d} + \bar{a}\bar{d} = \overline{\bar{a}\bar{d}} \cdot \overline{\bar{b}\bar{d}}$

Realizacja:



zadanie 6.

z siatki Karnaugh:

$$f(a,b,c,d) = \bar{a}cd + a\bar{b}\bar{c} + \bar{a}b\bar{d} =$$
$$= \underline{\bar{a}cd + a\bar{b}(\bar{c}+\bar{d})} =$$
$$= \underline{\bar{a}cd + a\bar{b} \cdot \overline{cd}} =$$
$$= \overline{\bar{a}\overline{cd} \cdot \overline{a\bar{b}}\,\overline{cd}}$$

Realizacja:

